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Soil moisture response to seasonal drought conditions and post-thinning forest structure

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Abstract

Prolonged drought conditions in semi-arid forests can lead to widespread vegetation stress and mortality. However, the distribution of these effects is not spatially uniform. We measured soil water potential at high spatial and temporal resolution using 112 sensors distributed across a ponderosa pine forest in northern Arizona, USA, during two abnormally dry years with below-average total precipitation. We used the data to assess the effects of fore-summer drought period on the timing, magnitude, and extent of drying throughout the top 100 cm of the soil profile. Additionally, we use high spatial resolution terrestrial lidar measurements of forest structure to develop relationships between soil drying and fine-scale forest structure. We find that increasing drought from 2019 to 2020 caused significantly earlier onset of soil dying at all depths (25, 50 and 100 cm) and more days below a critical drying threshold for ponderosa pine. Additionally, our results show that significantly drier soils are found in areas with higher stand-level basal area, canopy cover and tree density, and shorter trees. Our results from the unprecedented spatial and temporal resolution data suggest that tailored restoration thinning with specific tree density and size parameters can be used to increase and prolong the availability of deep soil water to trees during drought.

KEYWORDS
drought, forest health, forest moisture, forest mortality, forest thinning, lidar, soil moisture sensors, soil water potential

1 | INTRODUCTION

Persistent vegetation water stress can degrade the health and functioning of water-limited ecosystems (Allen et al., 2015; Porporato et al., 2001). While drought conditions are not abnormal in water-limited forest ecosystems, climate change-driven increases in air temperature and vapour pressure deficit are contributing to hotter and more frequent droughts as well as shifts in the average regional conditions leading to ecological drought (Bradford et al., 2020; Breshears et al., 2005; Vicente-Serrano et al., 2010, 2013; Zhang et al., 2021). When consistent soil moisture deficits occur in the vegetation root zone, especially during the growing season, the physiological processes controlling vegetation functioning, structure, and overall health are negatively impacted (Chapin, 1991; Chapin et al., 1987; Maherali & DeLucia, 2001; Williams et al., 2001). Adverse impacts to plants can accumulate during prolonged drought conditions (Adams et al., 2009; Palmer, 1965), and are further exacerbated when low soil moisture occurs simultaneously with high air temperatures (Breshears et al., 2018). Consequently, drought conditions can promote unsustainable levels of tree water stress and eventually contribute to widespread growth declines and tree mortality, especially in the semi-arid forests of the American Southwest (Allen et al., 2015; Clark et al., 2016; Ganey & Vojta, 2011; Koepke & Kolb, 2013; Mueller et al., 2005; Williams et al., 2010).

Spatio-temporal feedbacks drive differences in the movement and partitioning of water along the soil-vegetation-atmosphere continuum, helping determine the availability of soil moisture to vegetation across a landscape (Guswa et al., 2002; Koster et al., 2004; Seyfried et al., 2005; Thornthwaite, 1952). This high spatial and temporal variability in water cycling is enhanced in semi-arid ecosystems where high atmospheric moisture demand and infrequent, seasonally restricted precipitation contribute to lower baseline soil water levels and large fluctuations in soil moisture availability (Corradini, 2014; Loik et al., 2004). In some semi-arid ecosystems, up to 95% of annual precipitation inputs are used for vegetation transpiration and soil evaporation, with up to one third from the top $\sim$10 cm of the soil profile (Allen et al., 1998; Oki & Kanae, 2006; Stoy et al., 2019; Wang et al., 2014; Wei et al., 2017; Raz-Yaseef, Rotenberg, et al. 2010; Raz-Yaseef, Yakir, et al., 2010; Raz-Yaseef et al., 2012). In turn, these ecosystem-wide soil moisture dynamics promote variation in the spatial distribution of vegetation across the landscape (D'Odorico et al., 2007; Quevedo & Francés, 2008; Sandvig & Phillips, 2006; Snyder & Tartowski, 2006). Additionally, spatially variable soil moisture levels can translate into patterns in vegetation water stress and mortality at the landscape-scale (400 + ha) (Andrews et al., 2019; Goulden & Bales, 2019).

Landscape-scale spatial variability in vegetation water stress and mortality is enhanced at the fine-scale (<4 ha) by the distribution and orientation of forest patches, which also determines the amount of ground shading by forest canopy cover (Andrews et al., 2019; Raz-Yaseef, Rotenberg, et al. 2010; Raz-Yaseef, Yakir, et al., 2010; Teuling, 2005). In areas shaded by forest canopy, lower rates of soil evaporation and water yield increases have been observed (D'Odorico et al., 2007; Duff et al., 1997; Goeking & Tarboton, 2020; Qubaja et al., 2020; Sahin & Hall, 1996; Tyagi et al., 2013). At the fine-scale and within individual forest patches, these effects are more nuanced and differences in soil moisture levels can be driven by the structure and distribution of individual trees (Breshears et al., 1997; Gray et al., 2002; Lin et al., 1992). For example, within denser forest patches with higher canopy cover, increased canopy interception and tree water uptake result in lower soil moisture levels (Simonin et al., 2007). In contrast, higher soil moisture levels are often observed in less dense forest patches with lower canopy cover as well as within gaps directly adjacent to their north side (Gray et al., 2002; Raz-Yaseef, Rotenberg, et al. 2010; Raz-Yaseef, Yakir, et al., 2010).

In semi-arid ponderosa pine forests of the south-western United States, melt water from seasonal snowpack is a primary input to soil moisture throughout the soil profile. As with soil moisture, the spatial distribution of seasonal snowpack is governed by the size, shape, spacing, and structure of trees and tree groups (Belmonte et al., 2021; Davis et al., 1997; Dickerson-Lange et al., 2015; Donager et al., 2021; Essery et al., 2003, 2008; Lawler & Link, 2011; Molotch et al., 2009; Roth & Nolin, 2017; Sankey et al., 2015; Veatch et al., 2009). Forest stands with low or discontinuous canopy cover as well as interspaces adjacent to the north side of tree groups tend to have higher rates of snow accumulation, ablation, and persistence (Dickerson-Lange et al., 2017; Gottfried & Ffolliott, 1981; Revuelto et al., 2015). Differences in the spatial distribution of persistent snowpack are reflected in non-uniform soil water inputs during spring snowmelt (Newman et al., 2004). This effect is exacerbated by the high evapotranspiration rates during the fore-summer drought period, which significantly reduces the soil moisture levels in the upper soil horizon (<40 cm) (Brandes & Wilcox, 2000). While deep soil horizons provide the main source of water for mature trees of south-western pine forests, reductions in near-surface soil water inputs could reduce recharge rates as well as cause stress to shallow-rooted tree seedlings and herbaceous plants (Eggemeyer et al., 2009; Kerhoulas et al., 2013; Stahle et al., 2009). This has prompted forest managers to seriously consider the effects of landscape-scale soil water limitation and prioritize water management in forest ecosystems to reduce vegetation water stress and preserve productivity and resilience (Goeking & Tarboton, 2020; Grant et al., 2013). Quantifying the fine-scale patterns in forest structure-driven soil moisture variability can provide forest managers with actionable insight into the ecohydrological implications of thinning-based restoration practices.

Many forest ecosystems across the western United States have become overly dense and in turn increasingly vulnerable to widespread mortality from disease, drought, and catastrophic wildfire (Kolb et al., 1994; Larson & Churchill, 2012; Moore et al., 2004; Rautiainen et al., 2011; Reynolds et al., 2013; Swetnam & Betancourt, 2010). This has prompted the scientific community, government officials, and the public to increase funding for landscape-scale forest restoration (Allen et al., 2002; Covington & Moore, 2006; Fitch et al., 2018; Fulé, 2008; Schultz et al., 2012). A central component of forest restoration includes the selective thinning of overly dense stands to help reduce the risk of catastrophic wildfire, enhance wildlife habitat, promote vegetation health, and stabilize the water balance in treated forests. While the overarching goals of restoration treatments allude to promoting ecohydrological health and resiliency, they do not specifically address how selective thinning can be used to achieve this. As more forest is designated for restoration and the effects of climate warming continue to stress these ecosystems, a better understanding of the forest structure-soil moisture variability relationship is imperative for promoting landscape-scale forest resilience to global change droughts (Bradford et al., 2021).

Thinning treatments applied to south-western ponderosa pine forests have been shown to increase soil water availability and in turn reduce canopy water stress and increase carbon uptake of the remaining forest (Dore et al., 2012; Feeney et al., 2011; Sankey et al., 2020; Zausen et al., 2005). These earlier studies are important for assessing effects of specific thinning treatments on landscape-scale forest vulnerability to drought-induced growth declines and mortality, but they did not directly investigate soil moisture response to fine-scale variation in forest structure and how this differs along the soil profile (Gleason et al., 2017). Distinguishing such nuance will help tailor future restoration efforts to ensure that thinning utilizes specific tree structure and spatial patterns, potentially benefiting ecosystem water balance and promoting resilience to climate change effects.

Using a total of 112 soil water potential sensors, we provide a detailed, quantitative measurement of soil moisture availability and its persistence in response to drought and post-thinning forest structure conditions in a semi-arid forest. Specifically, we quantify soil moisture in the top 100 cm of the profile across a range of forest density conditions at a thinned restoration site over two consecutive years of 2019 and 2020. Both water years experienced below average precipitation, providing an opportunity to evaluate critical differences in soil moisture during high-stress years and particularly during the North American Monsoon season, which delivers much of the annual precipitation in northern Arizona (Figure 1). Total precipitation in both 2019 and 2020 were among the lowest on record in northern Arizona. Precipitation in water years of 2019 and 2020, were at 77% and 59% of the average precipitation, respectively. Figure 1 demonstrates the precipitation patterns in 2019 and 2020 compared to an average precipitation year 2018. In particular, summer 2019 and 2020 did not experience a typical North American Monsoon, which resulted in substantially lower precipitation than average.

Additionally, we assess how fine-scale forest structure components drive differences in the timing, magnitude, and amount of soil

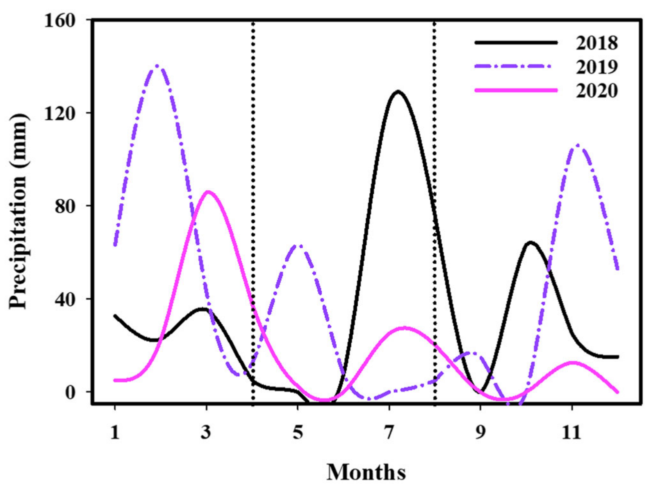

FIGURE 1 Precipitation patterns in 2019 and 2020 compared to an average precipitation year 2018 observed at the nearest (5.6 km away) National Oceanic and Atmospheric Administration (NOAA) climate data station (GHCND:USS0011P02S) to our study site. While a large snowstorm occurred in February 2019 and March 2020, both water years were exceptionally dry and especially during the fore-summer drought period from 1 April to early August, a focus period of our analysis (highlighted by vertical dotted lines). During this period, the North American Monsoon typically delivers much of the annual precipitation as shown for 2018 but delivered less than 50% of the average rainfall in 2019 and 2020

drying across soil depths during a critical drying period of the year following spring snowmelt. This important drying period is termed fore-summer drought period in this study (Figure 1). The fore-summer drought period starts on April 1 and ends at the onset of the North American Monsoon, which usually arrives in July (Figure 1). Soil moisture deficits experienced during the fore-summer drought period are believed to contribute to prolonged vegetation stress, reduced productivity, and enhanced forest mortality (Allen et al., 2015; Andrews et al., 2019), underscoring the potential for forest management treatments that increase soil moisture to promote forest health and resilience.

2 | MATERIALS AND METHODS

2.1 | Study site description

The study area is located in the Coconino National Forest about 6.5 km from the City of Flagstaff in northern Arizona, USA (12S 438346 N., 3901732 E. UTM). It includes 76 ha of forested land with flat topography (0%–10% slopes) and ranges in elevation between 2200 and 2275 m above sea level with an ephemeral watercourse running towards the southwest through the site (Figure 2). The region has a sub-humid climate with an average of 560 mm of precipitation and is characterized by strong seasonal trends including a winter snow, early summer drought, and late-summer monsoonal seasons (http://www.wrcc.dri.edu). Ponderosa pine *(Pinus ponderosa)* dominates both the region's and study area's vegetation, while the study area is punctuated by occasional Gambel oak *(Quercus gambelii)* and Rocky Mountain juniper *(Juniperus scopulorum)*. Arizona fescue (*Festuca arizonica*), mountain muhly (*Muhlenbergia montana*), mutton bluegrass (*Poa fendleriana*), bottlebrush squirreltail (*Sitanion hysterix*), and Buckbush (*Ceanothus fendleri*) comprise the understorey vegetation, which is typical of the region's ponderosa forest. While there was little variation in understorey species composition across our study site, total understory cover and abundance varied among plots. Specifically, plots with high tree density had lower understory vegetation cover compared to lower tree density plots. The last naturally occurring wildfire was recorded in the study area in 1876, but a prescribed fire was performed in 1976 that eliminated 63% of the smaller surface fuels and 69% of the larger woody surface fuels (up to 8 cm in diameter) (Dieterich, 1980; Sackett, 1980).

As a part of ongoing forest restoration efforts, a mechanical thinning treatment was implemented across the study area during 2017–2018 (Belmonte et al., 2019). This provided our study site with a 53-ha treated area of thinned forest adjoining a 23-ha untreated portion. The overarching goal of the restoration treatment was to reduce the risk of catastrophic wildfire by re-creating the less dense presettlement forest conditions and promote healthy overstory vegetation and the regeneration of understory vegetation (Allen et al., 2002). More specific treatment objectives included creating a wide size range of tree groups and interspaces, while promoting diversity in tree group shapes and their spatial arrangement across the treatment unit

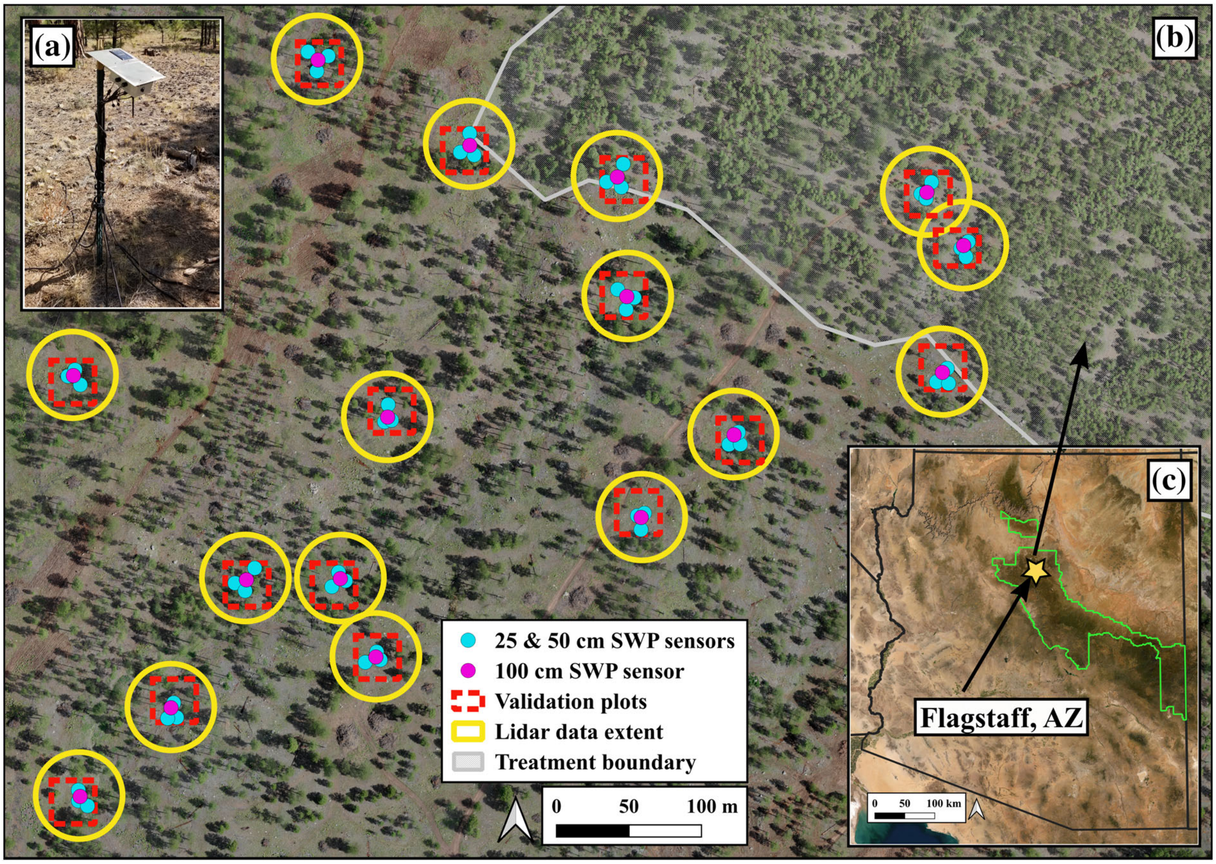

FIGURE 2 Study area and the location and extent of soil moisture and forest structure field data collection across the study site (b), which is within the larger 4FRI forest restoration area near Flagstaff, AZ, USA (c). All soil water potential sensors are connected to data loggers ($n = 16$) (a) located in areas with varying forest density. Forest structure data collection was centred on each data logger with overlapping lidar and validation datasets for accuracy assessment

(Belmonte et al., 2019; Larson & Churchill, 2012; Reynolds et al., 2013). The restoration treatment for the study site prescribed creating irregular tree groups, increasing overall interspace, retaining all non-ponderosa pine species, and significantly reducing the number of smaller ponderosa pine trees.

2.2 | Soil water potential data

Soil water potential (SWP) and soil temperature (ST) were measured using an array of dielectric water potential sensors ($n = 112$) installed at the centre of the forest structure data plots ($n = 16$) during the spring of 2018 (Figure 2). Each Meter Terros 21 (formerly Decagon MPS-6) sensor measures soil water potential (SWP) in kilopascals (kPa) with an accuracy of 0.1 kPa and soil temperature (ST) in degrees Celsius (°C) with an accuracy of 0.1°C (Decagon Devices). SWP values indicate the amount of pressure required to remove water from the soil. Overall, the more negative a SWP value, the more energy is required for plant roots to extract water from the soil and we did not partition water use by trees versus understory herbaceous vegetation. Each plot has two co-located sensors at depths of 25 and 50 cm at three different locations ($n = 6$ sensors per plot), and one additional sensor installed at depth of 100 cm (Figure 2). As a result, each plot has a total of seven sensors. Each sensor was programmed to record hourly measurements of both SWP and ST on a data logger located within each plot. The data from each plot was then wirelessly transmitted to a main data logger located at the centre of the study site, which compiled and transmitted the data over the cellular network to an offsite database (Yamamoto et al., 2010).

The full SWP dataset consists of hourly measurements throughout the year when sensors were operational (Figure 3). Here, we focus on the period lasting from April 1 through the end of the fore-summer drought in each year, since this is known to be a critical time for tree water availability. In this study, the fore-summer drought is defined as the period beginning when the spring snowmelt ceases and the soil is typically near full saturation (April 1), and lasts until the onset of the North American Monsoon season, which typically occurs in the mid-late summer. At our study site, the end of the fore-summer drought period was detected in SWP data at most sensors immediately following monsoonal precipitation that ranged between 5 and 10 mm at the beginning of August 2019 and at the end of July 2020 as well as early August 2020 (Figure 3). This estimate is based on precipitation data from the nearest weather station, located 5.6 km to the west, and is used to consistently define the end of the fore-summer drought

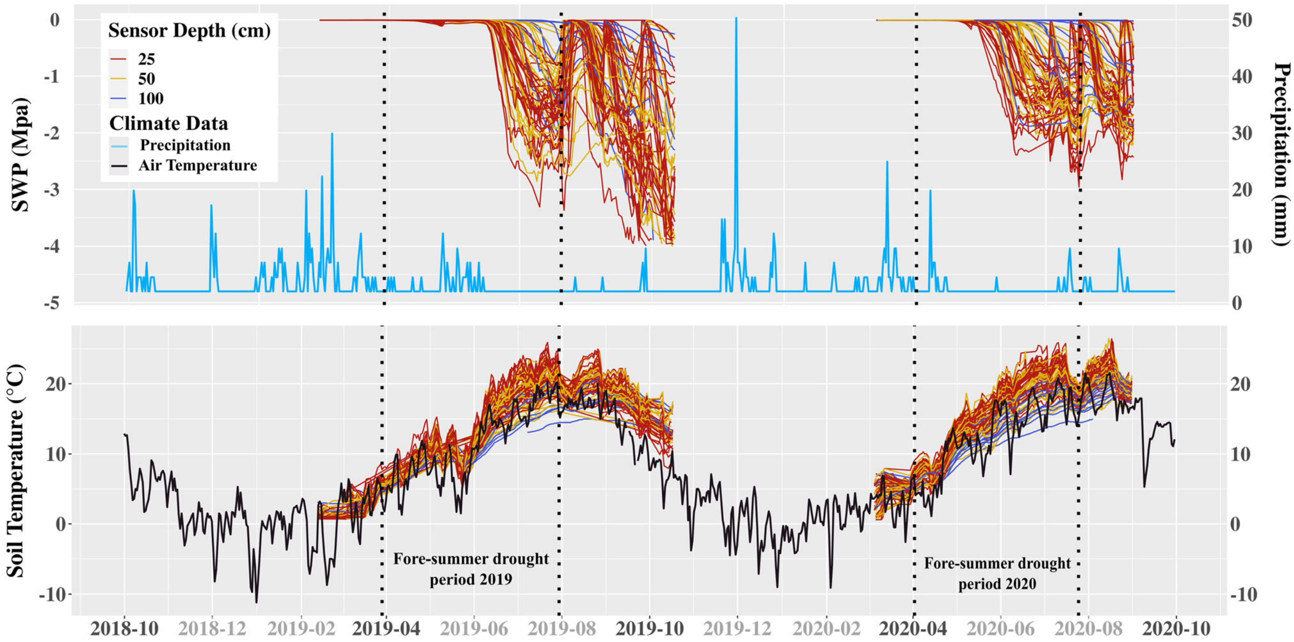

FIGURE 3 The complete soil moisture (SWP) and temperature (ST) time-series datasets for 2019 and 2020, with the respective fore-summer drought seasons highlighted with dashed lines. Precipitation (blue line in top panel) and air temperature (black line in bottom panel) data from the nearest (5.6 km away) National Oceanic and Atmospheric Administration (NOAA) climate data station (GHCND:USS0011P02S) are also shown for the full time frame. Based on the precipitation pattern, the end of the fore-summer drought period was detected in SWP data by 5-mm monsoonal precipitation at the beginning of August 2019 and 10-mm precipitation at the end of July 2020 and early August 2020

period across the study site, although soil moisture does not uniformly recover in varying forest density conditions. Sporadic equipment malfunction during the course of data collection resulted in slightly different numbers of sensors functioning each year. To mediate this issue, each sensor was characterized by its depth and a combination of localized forest structure metrics (described in forest structure section below). Each sensor's data was analysed to control for quality and quantity, and unrealistic values or large data gaps were thrown out entirely. The final time-series dataset consists of the average daily SWP value for each sensor throughout the fore-summer drought season (Figure 3).

The final time-series dataset was then used to generate three different metrics, as our key response variables, that quantify (1) the timing of soil drying onset, (2) magnitude of soil drying, and (3) overall moisture level at each sensor location. The first metric is termed onset, and it is a measure of when soil moisture drying occurred after 1 April. Onset corresponds to the number of days it took a sensor to begin a downward trajectory and fall below a threshold value of −0.1 MPa into a drying trend (more negative MPa values). The −0.1 MPa value is not significant to a physical property of soil drying. Rather, it is used as a proxy for the inflexion point, which marks a sensor's downward trend towards drying. Saturated soils in our data show 0 MPa value with some daily and hourly fluctuations around that value. It was, therefore, important to exclude these small fluctuations and mark a real downward trend with the −0.1 MPa value. The second metric measures the number of days below −1.0 MPa, which is a critical drying threshold (CDT) causing vegetation stress in ponderosa pine forests (Breshears et al., 1997; Gaylord et al., 2007; Gyenge et al., 2002; Sala et al., 2005). The final metric is the average soil moisture or SWP value (MPa), termed meanSWP, experienced by a sensor starting from the day it crosses the onset threshold and stopping at the end of the fore-summer drought period.

2.3 | Forest structure metrics

Forest structure was estimated for each plot using terrestrial laser scanner (TLS) point cloud data collected from 0.28 ha plots ($n = 16$; 4.48 ha) centred on each SWP data logger (Figure 4). The TLS data were collected during the summer of 2019 using a Leica Geosystems BLK360 Imaging Laser Scanner (Leica Geosystems AG, 2020). Each TLS plot included three or more scans that were merged together to ensure full coverage and georeferenced with fixed reference targets on adjustable tripods at 1 m above ground. Targets were four 50 × 50 cm reflective reference panels strategically located within each plot for optimal scan-to-scan visibility (Donager et al., 2018). The location of each reference target was recorded at its centre point (1 m AGL) and corrected using the Trimble GeoXH handheld GPS unit and GPS Pathfinder Office software. TLS data post-processing was performed using CloudCompare and consisted of creating a single georeferenced point cloud for each plot. Each scan was aligned with another using the four ground target centre points as common points, until all scans were merged into a single point cloud. Then, each point cloud was georeferenced using the GPS coordinates of the target; the

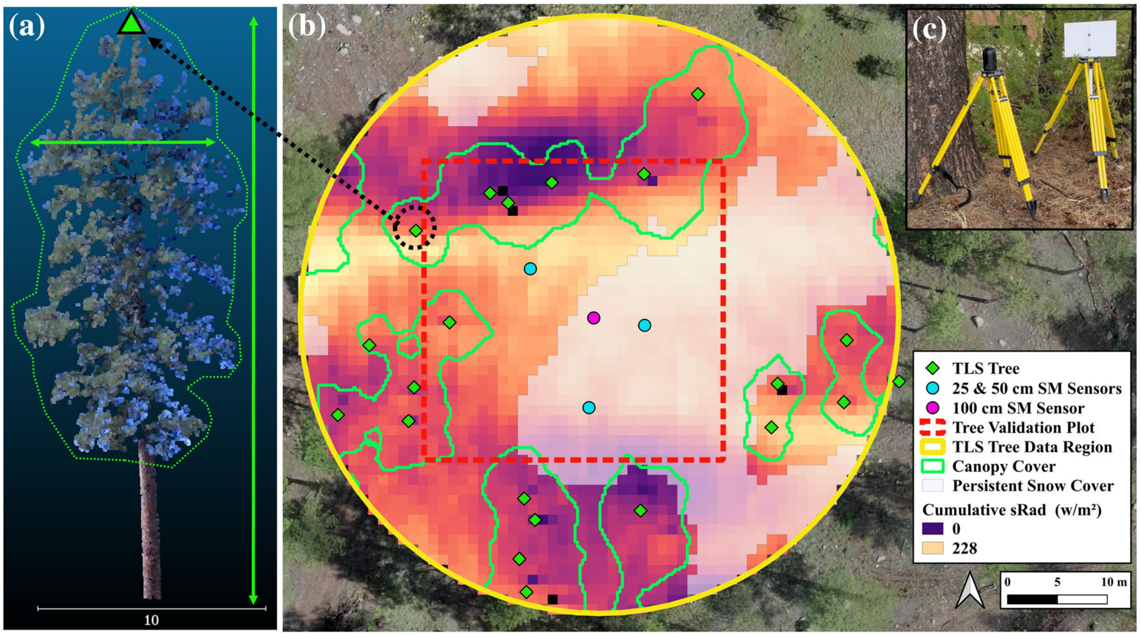

FIGURE 4 Forest structure metrics were calculated using plot-level lidar data collected at each soil moisture data logger location ($n = 16$). Additionally, validation measurements were collected in 30 × 30-m plots to assess the accuracy of the lidar-derived forest structure estimates. Panel (a) depicts a segmented tree from the lidar point cloud data, while panel (b) provides an illustration of all the lidar-derived forest structure metrics estimated at a single plot. Panel (c) shows the terrestrial laser scanner (TLS) instrument and one of the ground targets used during data collection

overall *X*, *Y* and *Z* positional accuracy of the point clouds ($n = 16$) is 0.46 m. Finally, the average point spacing of all point clouds was 0.01 m (ranging 0.025–0.046 m), and they were subsampled to 0.03 m to ensure consistent point spacing across scans.

To estimate forest structure metrics, the TLS point clouds were first post-processed and classified into ground/non-ground points. Then an individual tree segmentation was performed using the Li et al. (2012) algorithm in the lidR package (Roussel et al., 2017) in RStudio (R-Studio Team, 2015). The resulting segmented point clouds were then used to estimate each tree's location (*X*, *Y* in UTM 12 N m), crown height (m), and average of its widest and narrowest crown diameters (m) using the rLiDAR package in R-Studio (Mohan et al., 2017). Finally, the diameter at breast height (DBH in cm) was estimated at 1.37 m above the ground level for each segmented tree using the TreeLS package in RStudio (de Conto et al., 2017). The TLS point cloud data were validated using field-based measurements taken from 137 trees located within 30 × 30-m validation plots ($n = 16$; 1.44 ha) and were also centred at SWP data logger locations. Using the Trimble GeoXH handheld GPS unit with an attached laser range finder module, each tree's geographic position (*X*, *Y* and *Z* coordinates), crown height and diameter were measured and differentially corrected in GPS Pathfinder Office. Additionally, each tree's diameter at breast height in cm was measured using a diameter tape at a height of 1.37 m above the ground level.

To assess the differences in sensor-level soil moisture related to forest structure, a set of forest structure summaries were calculated specific to each sensor's location (Figure 4). To accomplish this, trees were selected from each TLS point cloud within a 10 m radius footprint unique to each sensor. Once the trees were selected, their crown height, diameter at breast height, and crown volume were averaged. Next, the stand basal area (m^2/ha), trees per hectare, and percent canopy cover were calculated across the 10 m radius footprint. Separately, the total incoming ground-surface solar radiation (w/m^2) was calculated for the dates comprising the fore-summer drought period to quantify the effects of forest canopy shading. This was accomplished using a ray-tracing algorithm relying on the TLS point cloud data to calculate the accumulated hourly solar radiation based on the high-resolution structure of each tree (Seyednasrollah et al., 2013; Seyednasrollah & Kumar, 2014). Additionally, the total area (m^2) of persistent snow cover was calculated during the winter months across the plot to quantify the potentially lingering effects of forest-structure-driven snow persistence from the winter season (Belmonte et al., 2021). This was accomplished using data from UAV image-derived time-series analysis of snow cover at the same study site in 2018 and 2019 (Belmonte et al., 2021). Finally, after each sensor's forest structure metrics were calculated, each metric was categorized into three groups based on the statistical distribution of its values: low, medium and high.

2.4 | Data analysis

Differences in the soil moisture metrics (onset, critical drying threshold, and meanSWP) for each year (2019 and 2020), soil depth

(25, 50 and 100 cm), and forest structure metrics (crown height, DBH, crown volume, basal area, trees per hectare, canopy cover, solar radiation and snow cover) were assessed using a series of one-way ANOVA tests. First, differences in soil moisture response were compared between the 2019 and 2020 fore-summer drought seasons using data from sensors across all depths. Secondly, data from both years was used to test for soil moisture differences among soil depths. Finally, data from both years was used to test for significant differences in soil moisture in response, parsed by depth, to forest structure metrics, which were grouped into low, medium, and high conditions. Assumptions of equality in depth, year, and forest structure group-level variances and normality in residuals were tested using Bartlett's and Shapiro–Wilk tests, respectively. Additionally, inspection of group-level histograms and boxplots were used to identify any indications of non-normality. In the case of significant differences among soil depth groups, Tukey's HSD post hoc test was performed to assess group-level differences.

3 | RESULTS

3.1 | Soil moisture data summary

After April 1, the two consecutive fore-summer drought seasons spanned 120 days and 115 days in 2019 and 2020, respectively. The 2019 fore-summer drought season ended on 30 July 2019, while in 2020 it ended on 25 July 2020, with the first soil-wetting rainstorm aligning with increased SWP levels and decreases in both soil and air temperatures (Figure 3). The SWP measurements from the 2019 fore-summer drought season consisted of continuous hourly time-series data from 66 total sensors: 31 sensors at 25 cm, 26 sensors at 50 cm, and 9 sensors at 100 cm. The 2020 fore-summer drought time-series data included 66 total sensors: 30 sensors at 25 cm, 24 sensors at 50 cm, and 12 sensors at 100 cm. Figure 5 illustrates the mean SWP values summarized by year and depth and includes both the onset and critical drying thresholds to illustrate the differences in these metrics by year and depth.

3.2 | Soil moisture differences between years

Combining data from all sensors, we observed a significant difference in onset between the 2019 ($M = 177.1$, $SD = 17.8$) and 2020 ($M = 158.2$, $SD = 19.1$) fore-summer drought seasons ($F[1,130] = 78.8$, $p = 4.6e-15$). On average, onset started 18.8 days ($p < 0.001$) earlier in 2020 than in 2019. There was also a significant difference in CDT between the 2019 ($M = 19.6$, $SD = 13.5$) and 2020 ($M = 27.3$, $SD = 19.1$) fore-summer drought seasons ($F[1,130] = 10.5$, $p = 0.001$). On average, there were 7.7 additional days in 2020 ($p = 0.003$) spent below the CDT than in 2019. There were no significant differences in mean SWP between the 2019 and 2020 seasons.

3.3 | Soil moisture differences among depths

When considering data by soil depth, there were significant differences in the onset between sensors at 25 cm ($M = 157.9$, $SD = 14.4$), 50 cm ($M = 168.3$, $SD = 13.8$), and 100 cm ($M = 194.2$, $SD = 26.4$) (Figure 6). Sensors at 25 cm had an onset 10.1 days ($p = 0.0002$) and

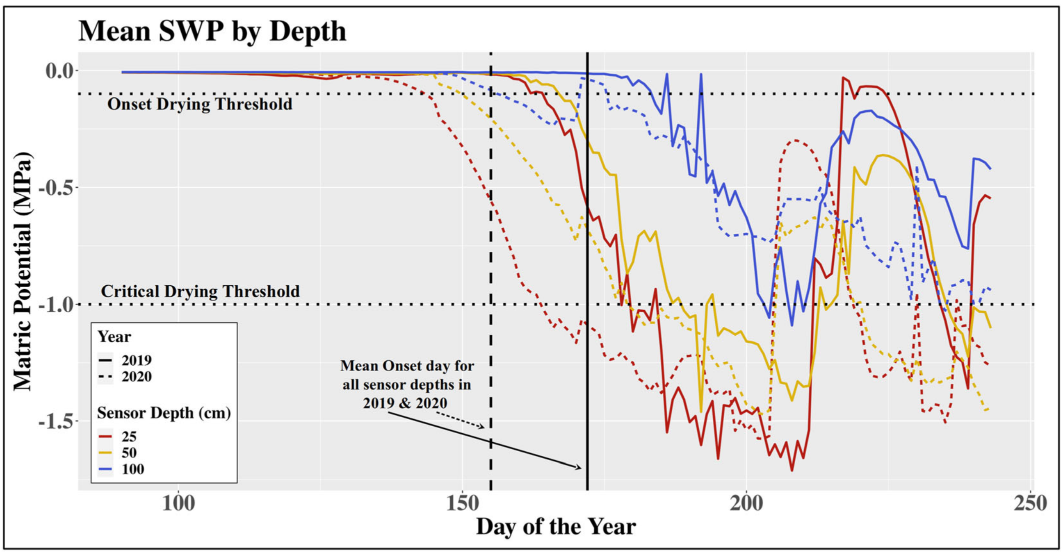

FIGURE 5 Soil moisture time-series data from 2019 and 2020 fore-summer drought seasons. 1 April, the start of the fore-summer drought period occurred on day 91 and day 92 for 2019 and 2020, respectively (Julian days). Each line shows the mean daily soil water potential (SWP) measurement (coloured by depth) and illustrates the soil 'drying down' between spring snow melt (1 April) and late summer monsoon. Rain events can be observed when a curve sharply turns upward towards 0 MPa, which indicates soil becoming more saturated. The critical drying threshold (CDT) depicted at −1.0 MPa is the approximate value at which ponderosa pine trees begin to experience moisture-related stress

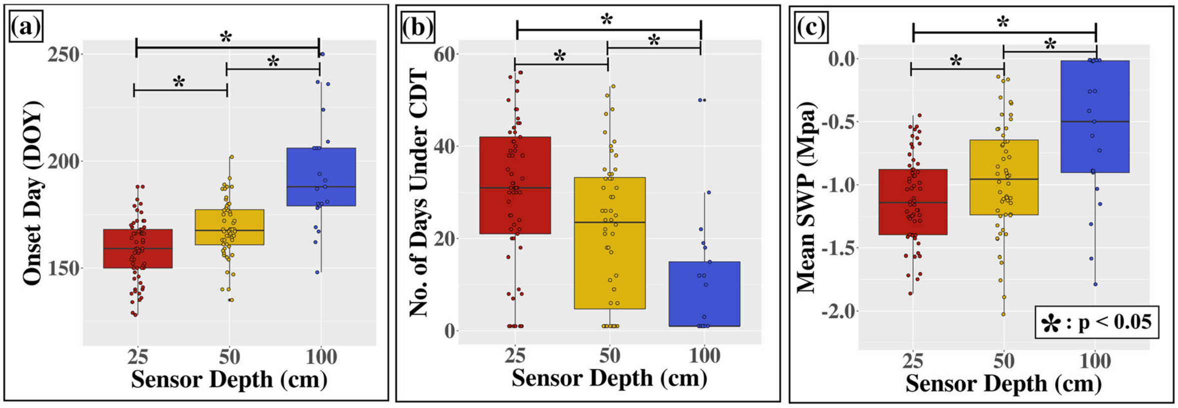

FIGURE 6 Mean differences in each soil moisture metric by depth. Panel (a) shows significant differences for onset day (DOY) among soil depths. Panel (b) shows significant differences in the number of days spent under the critical drying threshold (CDT), and panel (c) shows differences in the mean SWP (MPa). Significant differences (marked with *) were found for each depth-level comparison. Boxes indicate interquartile range and whiskers show 1.5 times interquartile range

37.7 days ($p = 0.00$) earlier than sensors at 50 and 100 cm, respectively. Sensors at 50 cm had an onset 27.5 days ($p < 0.001$) earlier than those at 100 cm. There were significant differences in the number of days spent below the CDT between sensors at 25 cm (M = 29.9, SD = 15.8), 50 cm (M = 21.4, SD = 15.9) and 100 cm (M = 9.6, SD = 12.7) (Figure 6). Sensors at 25 cm spent an additional 8.5 days ($p = 0.008$) and 20.9 days ($p = 0.0005$) under the critical drying threshold than sensors at 50 cm and 100 cm, respectively. Sensors at 50 cm spent an additional 12.5 days ($p = 0.004$) below the critical drying threshold than those at 100 cm. Additionally, there were significant differences in the mean SWP values between sensors at 25 cm (M = −1.1 MPa, SD = 0.37), 50 cm (M = −0.95 MPa, SD = 0.44) and 100 cm (M = −0.59 MPa, SD = 0.56), with sensors at 100 cm being 0.54 and 0.36 MPa wetter than those at 25 and 50 cm, respectively (Figure 6).

3.4 | Forest structure - soil moisture differences

Forest structure metrics were derived using trees segmented from the lidar point cloud, which correctly identified 98% ($n = 1280$) of the trees from within the field-measured validation plots ($n = 16$ validation plots, totaling 5.17 ha). An accuracy assessment compared a subset of tree metrics between the lidar-derived and field-measured trees including tree crown height (m), crown diameter (m), DBH (cm) and geographic location (*X,Y* in UTM 12 N m). Relationships between the lidar-derived and field-measured trees were generally strong, with a $R^2 = 0.95$ for tree crown height. A full summary of the accuracy assessment process and forest structure metric relationships are reported in Belmonte et al. (2021). Forest structure summaries were then calculated specific to each SWP sensor location. Figure 7 illustrates how forest structure was sampled by the entire sensor collection.

When considering the onset day at 25 cm, we observed significant differences related to tree crown height ($F[2,57] = 6.8$, $p = 0.002$), diameter at breast height ($F[2,57] = 2.9$, $p = 0.05$), solar radiation ($F[2,57] = 7.8$, $p = 0.0009$), basal area ($F[2,57] = 11.6$, $p = 0.0006$), trees per hectare ($F[2,57] = 11.5$, $p = 0.0008$) and canopy cover ($F[2,57] = 22.1$, $p = 9.77e{-}06$), with significant group-level differences present for crown height, basal area, trees per hectare and canopy cover (Table 1). At 50 cm, there were significant differences in onset day related to tree crown height ($F[2,48] = 15.8$, $p = 5.89e{-}06$), basal area ($F[2,48] = 23.1$, $p = 4.72e{-}05$), trees per hectare ($F[2,48] = 13.8$, $p = 0.0005$), snow cover ($F[2,48] = 4.7$, $p = 0.01$), and canopy cover ($F[2,46] = 7.3$, $p = 0.001$), with significant group-level differences in tree crown height, basal area, trees per hectare and canopy cover (Table 1). There were no significant differences in onset day at 100 cm due to forest structure.

When considering the number of days spent below the critical drying threshold at 25 cm, we observed significant differences related to tree crown height ($F[2,57] = 9.6$, $p = 0.002$), diameter at breast height ($F[2,57] = 5.4$, $p = 0.006$), and canopy cover ($F[2,57] = 7.5$, $p = 0.0013$), with significant differences among groups within tree crown height and diameter at breast height (Table 1). At 50 cm, there were significant differences related to tree crown height ($F[2,46] = 17.9$, $p = 1.5e{-}06$), snow cover ($F[2,46] = 10.4$, $p = 0.0001$) and basal area ($F[2,46] = 14.9$, $p = 8.7e{-}06$), with significant differences among tree crown height and snow cover groups (Table 1). Again, there were no significant differences in the number days below the CDT at 100 cm related to forest structure.

Finally, when considering the mean SWP values, sensors at 25 cm had significant differences related to tree crown height ($F[2,57] = 3.9$, $p = 0.001$) and diameter at breast height ($F[2,57] = 4.5$, $p = 0.01$) (Table 1). At 50 cm, the significant differences in mean SWP values were related to tree crown height ($F[2,46] = 7.3$, $p = 0.001$) and basal area ($F[2,46] = 4.5$, $p = 0.01$), with significant differences among groups for tree crown height and basal area (Table 1). At 100 cm, there were no significant differences in mean SWP values.

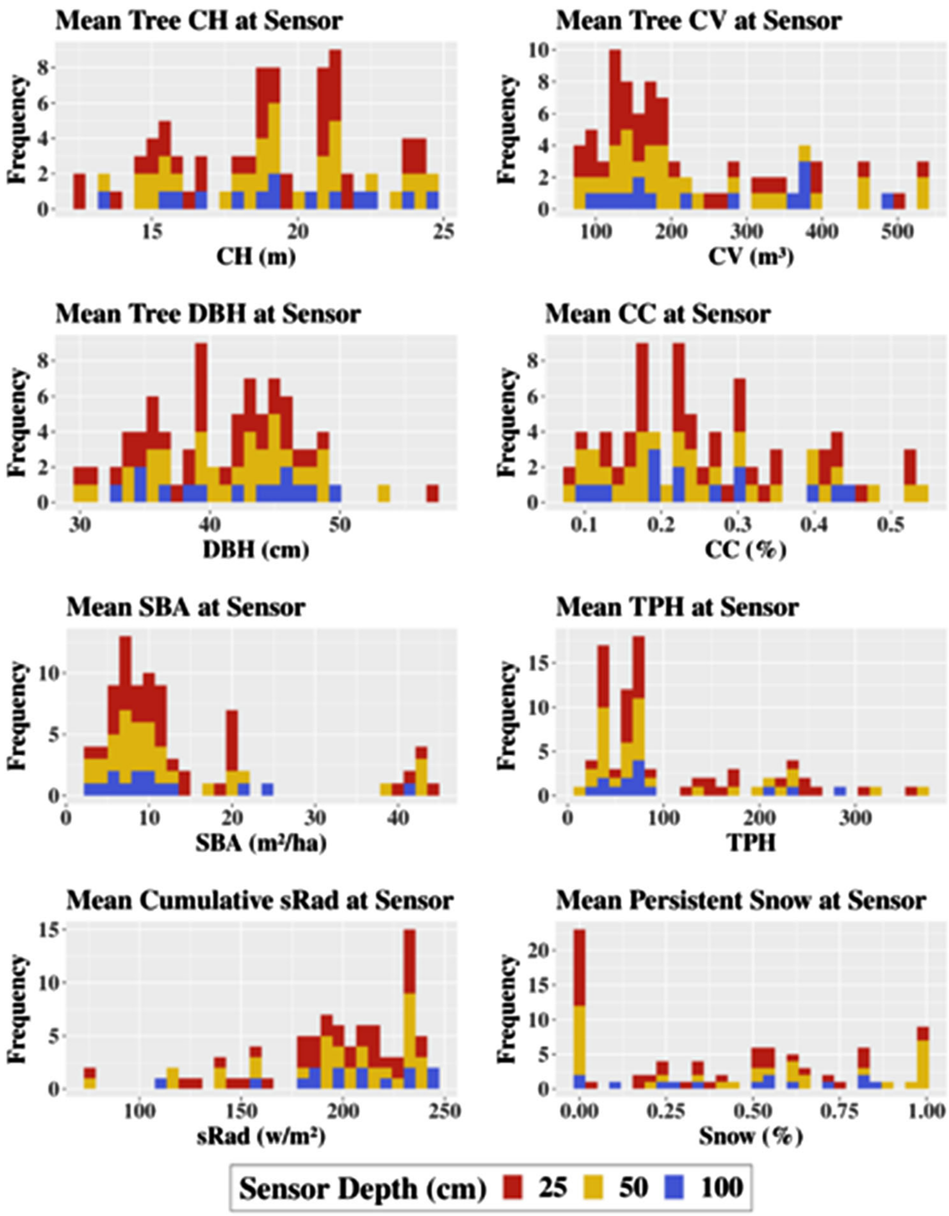

FIGURE 7 Histograms of the forest structure metric summaries calculated at each sensor location ($n = 84$), with sensor depths differentiated by colour. Key indicators of the post-treatment forest structure are apparent in the distributions of mean canopy cover (CC), basal area (SBA), and trees per hectare (TPH), all of which reflect the overall lower forest density conditions across the study site compared to the conditions observed prior to the restoration thinning project (Belmonte et al., 2019)

4 | DISCUSSION

In this study, we assess soil drying during two consecutive fore-summer drought periods in the midst of regional multi-year drought (Chikamoto et al., 2017). The timing, magnitude and amount of soil drying are critical to assessing the severity of seasonal drought and provide context for the impacts of multi-year drought. We also examine relationships between soil moisture and forest structure to improve our understanding of local-scale soil moisture variability. Our results show three important relationships between fine-scale patterns in soil drying and post-thinning forest structure: 1) significant differences in drying trends between two fore-summer drought seasons, 2) significant differences in soil drying trends across soil depths, and 3) significant differences in soil drying based on forest structure. Our findings provide insight into the potential impacts that landscape-scale restoration can have on soil moisture persistence during the region's driest period of the year.

4.1 | Multi-year drought impact

Our study site experienced below average precipitation (77% and 59% of normal, in 2019 and 2020, respectively) in both water years preceding each fore-summer drought period, although the length of the fore-summer drought periods was roughly equal in 2019 and 2020 (120 vs. 115 days, respectively). These notably dry conditions provided an excellent opportunity to study both the general response of SWP to limited water resources as well as to forest structure because it provided a clear overall drying signal. We first show that soil drying behaviour was significantly different between the fore-summer drought seasons of 2019 and 2020 despite no significant

TABLE 1 Summary of the significant Tukey's post hoc group-level comparisons for the combinations of SM and FS metrics parsed by sensor depth

SM metric	Sensor depth	FS metric	Tukeys post hoc comparisons	Mean difference ($p < 0.05$)
Onset day	25 cm	CH	Tall versus short trees	+13.2 days
		BA	High versus low density	−10.5 days
			High versus medium density	−14.9 days
		TPH	High versus medium density	−12.2 days
		CC	High versus medium cover	−15.5 days
	50 cm	CH	Medium versus short trees	+13.6 days
			Tall trees versus short trees	+23.6 days
			Tall trees versus medium trees	+10.1 days
		BA	High versus low density	−12.6 days
			High versus medium density	−18.6 days
		TPH	High versus low density	−14.5 days
			High versus medium density	−13.8 days
		CC	High versus medium cover	−12.7 days
CDT days	25 cm	CH	Tall versus short trees	−19.2 days
			Tall versus medium trees	−14.6 days
		DBH	Large versus small trees	+11.3 days
			Large versus medium trees	+15.8 days
	50 cm	CH	Tall versus short trees	−28.7 days
			Tall versus medium trees	−11.7 days
			Medium versus short trees	−17.1 days
		BA	High versus low density	+12.9 days
			High versus medium density	+26.3 days
		TPH	High versus low density	+18.1 days
			High versus medium density	+17.7 days
		CC	High versus medium cover	+16.9 days
		SC	Medium versus low snow	+19.8 days
			High versus low snow	+15.1 days
Mean SWP	25 cm	CH	Tall trees versus short trees	+0.33 Mpa
			Tall trees versus medium trees	+0.29 Mpa
		DBH	Large versus small trees	−0.29 Mpa
			Large versus medium trees	−0.28 Mpa
		sRad	High versus low exposure	−0.28 Mpa
			High versus med exposure	−0.44 Mpa
		CC	Medium versus low cover	+0.36 Mpa
	50 cm	CH	Tall versus short trees	+0.59 Mpa
			Tall versus medium trees	+0.41 Mpa
		BA	Medium versus low density	+0.41 Mpa
		SC	Medium versus low snow	−0.52 Mpa
			High versus low snow	−0.69 Mpa

Note: Forest metric group value ranges are as follows: Tree crown height (CH), short (<15 m), medium (15–20 m), tall (>20 m); diameter at breast height (DBH) small (<30 cm), medium (30–45 cm), large (>45 cm); basal area (BA) low (<8 m^2/ha), medium (8–16 m^2/ha), high (>16 m^2/ha); trees per hectare (TPH) low (<60), medium (60–130), high (>130); canopy cover (CC) low (<15%), medium (15%–30%), high (>30%), solar radiation (sRad) low (<180 w/m^2), medium (180–220 w/m^2), high (>220 w/m^2); snow cover (SC) low (<25%), medium (25%–60%) and high (>60%).

differences in overall measured SWP levels and length of fore-summer drought period. This is reflected in both the timing of soil drying onset, which occurred ~19 days earlier in 2020 than in 2019, as well as ~8 more days spent below the critical drying threshold (CDT) in 2020 than in 2019 (a 39% increase in dry days compared to 2019). The below average precipitation in 2019 and its continued trend likely set the stage for the significantly lower soil moisture observed in 2020. Additionally, after the 2019 fore-summer drought period, a further soil drying trend was observed with many sensors recording SWP values below the critical drying threshold (Figure 3). Interestingly, the similar overall mean SWP values between the fore-summer drought periods in the 2 years hints at an underlying and consistently low soil moisture availability when considering the entire soil profile, likely caused by an ongoing multi-decadal megadrought in the southwestern United States (Williams et al., 2020). These results have important implications for local and regional forest health and specifically water stress for trees. Continued multi-year meteorological drought conditions can exacerbate soil drying in the deeper and more stable portions of the soil profile, increasing the likelihood for tree mortality from prolonged soil moisture deficits (Breshears et al., 2018; Fettig et al., 2019; Goulden & Bales, 2019).

Next, we show that differences in soil moisture metrics among soil depths occur regardless of year and forest density conditions (Figure 5). More specifically and consistent with previous studies (Breshears et al., 1997; Goulden & Bales, 2019), soil moisture measurements from shallower sensors exhibited significantly earlier soil drying onset, more days spent below the CDT, and lower mean SWP values compared to the deeper sensors (Figure 6). Our results underscore overall wetter conditions and greater resistance to drying at increasing soil depth and this was consistent despite the ongoing meteorological drought conditions and differences in forest structure. This is shown in the overall significantly wetter conditions at 100 cm compared to more shallow depths; some of the sensors at 100 cm never entered a drying phase or spent time under the CDT. Previous research has consistently showed that soil drying occurs more rapidly and completely at and near the surface due to increased exposure to solar radiation and the greater density of roots (Capehart & Carlson, 1997; Huang et al., 2018; Martinez et al., 2008). This was observed in nearly all of the sensors at 25 cm having an early onset, spending more time below the CDT, and exhibiting more complete drying overall compared to the deeper depths (Figure 3). Importantly, we show that soil moisture at 50 cm can reflect both the extreme drying trends observed at 25 cm as well as the underlying stability characteristic at 100 cm. Given the multi-year drought conditions experienced during both fore-summer drought periods, these effects were likely exacerbated by the range in forest structure and density conditions present across the study site from the previously implemented restoration thinning project (Belmonte et al., 2021).

4.2 | Forest structure effects

Our results show soil moisture at a local scale was significantly and consistently impacted by five specific forest structure metrics: mean tree crown height, mean tree diameter at breast height, basal area, canopy cover, and trees per hectare. Importantly, all the effects of forest structure on soil moisture metrics were observed at 25 and 50 cm, with no significant effects related to soil moisture at 100 cm (Figures 7–9). At both 25 and 50 cm, the forest structure metrics related to density conditions (basal area, canopy cover, and trees per hectare) appeared most frequently as significant variables influencing soil moisture (Table 1). Higher levels of basal area, canopy cover, and trees per hectare all translated to an earlier onset of soil drying and more days spent below the CDT at 25 and 50 cm, while the effects on mean SWP values were more nuanced. Interestingly, higher basal area and canopy cover values translated to wetter mean SWP levels at shallow sensors. However, these significant differences were related to only 'medium' and 'low' categories and can possibly be explained by conditions in the medium basal area and canopy cover groups providing just enough shading to harbour shallower soil moisture content via reduced soil evaporation. While the soil moisture response at 100 cm did not show significant effects related to forest structure metrics, there were clear trends similar to these at 25 and 50 cm. However, these trends were observed towards the most extreme ends of the forest density gradient, where less dense forest exhibited wetter overall soil conditions than denser forest (Figure 10). Additionally, the higher average tree crown height values also translated to overall wetter soil moisture conditions at 100 cm. This hints at the potential for managing deeper soil moisture via restoration thinning practices.

Tree crown height is the one metric related to individual tree structure that consistently influenced soil moisture response. Specifically, sensors near taller trees had a later onset of soil drying, spent less days below the CDT and had greater overall soil moisture levels (Table 1). Taller trees have larger shadow footprints, and thus provide more ground shading during the fore-summer drought period. This suggests an important benefit to soil moisture persistence from 'strategic' ground shading. This reinforces not only the complex relationships between soil moisture and forest/tree structure but also the potential in tailoring restoration thinning to promote soil moisture persistence.

Persistent snow cover levels also had a significant effect on soil moisture response at 50 cm, with more days below the CDT and overall lower mean SWP levels at locations with higher amounts of persistent snow cover throughout the winter season. Counter intuitively, this points to overall drier soil conditions during the fore-summer period in locations with greater amounts of persistent snow during the winter months. Mechanisms for drier soil beneath more persistent snow patches in the spring include greater sublimation driven by higher temperature, which would reduce snow melt into the soil. Another contributing factor might be a difference in net seasonal solar radiation at these locations in the winter months versus the early summer months. Our previous observations at this study site show that the primary drivers of persistent snow cover during the winter months were associated with moderate levels of forest density (Belmonte et al., 2021). While this overlaps with our current findings related to soil moisture levels, we found no direct linkages between persistent snow cover during the winter months and fore-summer soil moisture via specific forest structure conditions. Instead, our results point to a disconnection between forest-structure-driven trends observed in persistent snow cover during the winter months and those observed in soil drying during the fore-summer drought period.

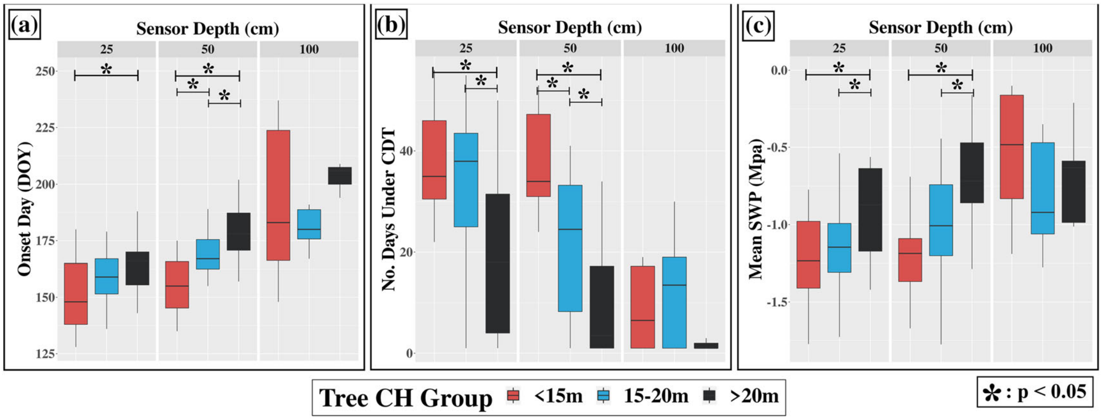

FIGURE 8 Differences in onset day (panel a), number of days below the critical drying threshold (panel B), and mean SWP (panel C) related to tree crown height (CH). Boxes indicate interquartile range and whiskers show 1.5 times interquartile range. Significant differences in Tukey's pairwise comparisons are shown by the starred bars. Mean tree crown height was the most prevalent predictor of significant differences in soil moisture metrics across all soil depths

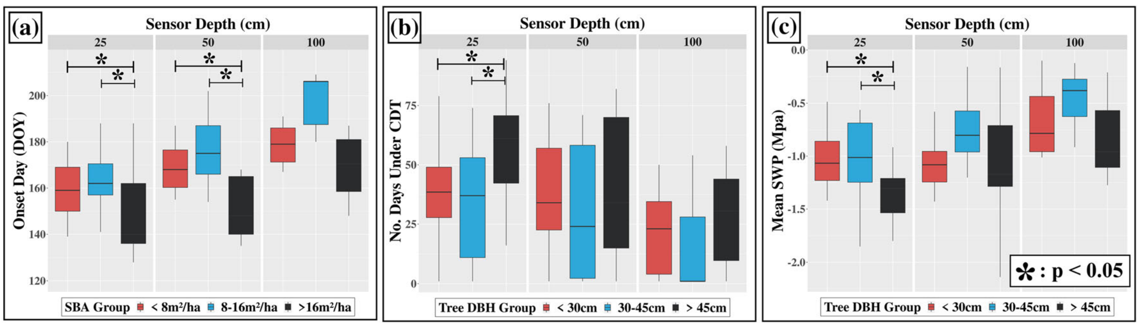

FIGURE 9 Notable forest structure-driven differences in each soil moisture metric. Boxes indicate interquartile range and whiskers show 1.5 times interquartile range. For onset day (a), basal area (BA) had significant pairwise differences for sensors at 25 and 50 cm, with higher BA translating to significantly earlier onset day. Additionally, there were significant pairwise differences in tree diameter at breast height related to the number of days below the critical drying threshold (b) and mean SWP (c) for sensors at 25 cm. Here, trees with larger diameter at breast height contributed to more days spent under the threshold and lower (drier) overall SWP values

This disconnect might be related to differences between snow persistence and total snow water equivalent, and underscores an important knowledge gap related to the temporal scale of ecosystem moisture inputs as well as the larger ecohydrological impacts caused by forest structure. Addressing this gap will be vital to enable forest management for moisture regardless of the season.

4.3 | Management implications

Forest managers will increasingly need to address the nexus of ecosystem water inputs and forest structure as restoration thinning projects continue to be implemented across the dry forests of the western United States and regional climates trend towards hotter and drier conditions. Our results suggest that managers can target soil moisture levels at 50 cm to most effectively impact and monitor forest ecohydrological cycles. Soil drying at 25 cm was rapid and highly variable in response to weather and forest structure conditions, ultimately shedding little insight into the soil moisture trends deeper in the profile. Furthermore, soil moisture at 25 cm is documented to be influenced by understory herbaceous vegetation evapotranspiration, which can be substantial (Springer et al., 2006). Total understory cover and abundance at our study site varied among plots. Specifically, plots with high tree density had lower understory vegetation cover compared to lower tree density plots. This variation might have contributed to the observed differences in soil moisture availability at 25 cm. Our study did not partition soil water use by trees versus understory vegetation and, therefore, cannot separate the understory vegetation contribution towards the observed differences in soil moisture at 25 cm. Future research should address this partitioning to determine how much of the soil moisture variability at 25 cm is influenced by understory vegetation versus forest structure during extended drought periods.

Soil moisture at 50 cm reflected both more stability yet also significant responsiveness to forest structure conditions. Our results indicate that forest managers can continue to implement restoration

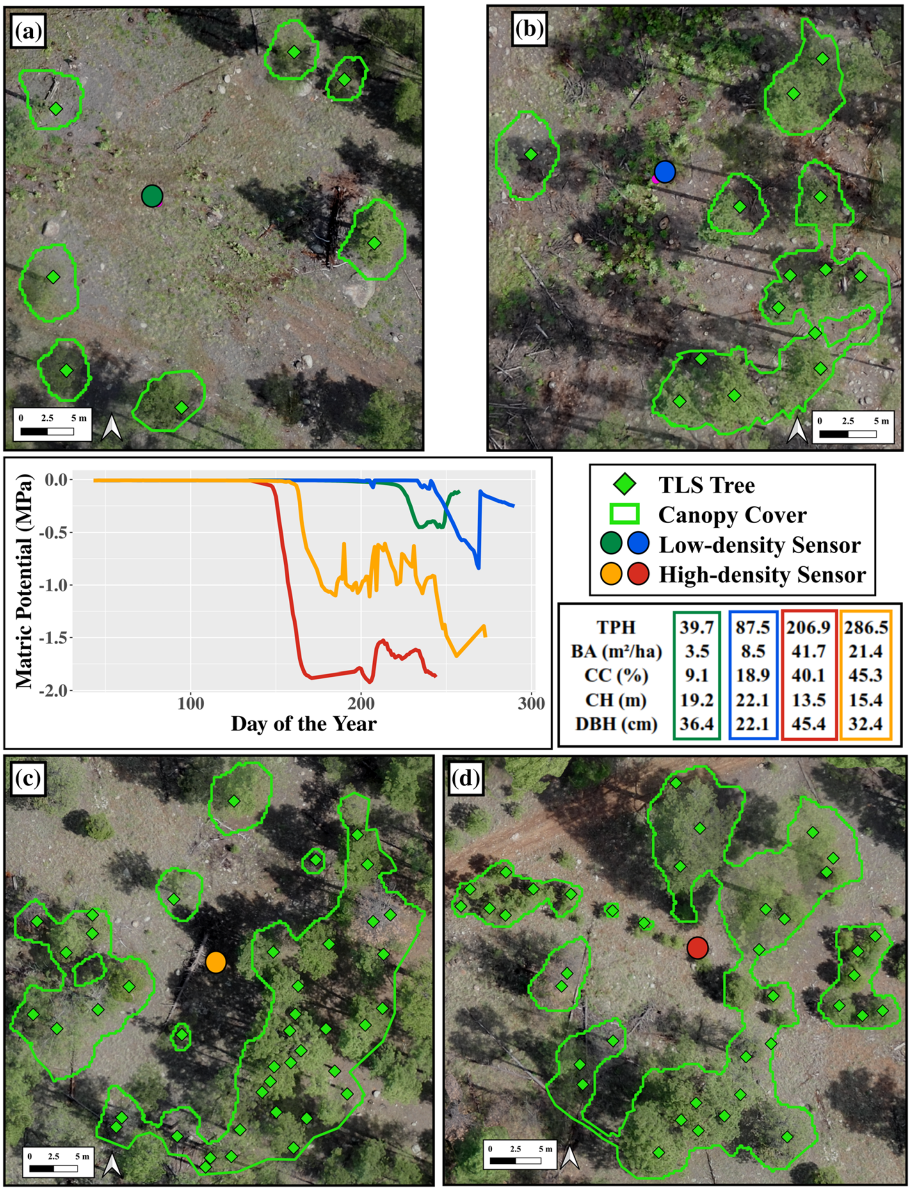

FIGURE 10 Illustration of four different 100 cm soil matric potential (MPa) time-series datasets: Two from plots with the lowest forest density conditions (a and b) and two from the highest forest density conditions (b and c). Soil matric potential data (middle panel) was averaged across both fore-summer drought seasons for each sensor to illustrate the soil moisture response at that location. Trees per hectare (TPH), basal area (BA), canopy cover (CC), average tree crown height (CH) and average diameter at breast height (DBH) were summarized across the 20-m radius area centred on each sensor

prescriptions focused on density reductions and anticipate less severe soil drying at 50 cm after restoration treatments (Table 1). Continued monitoring of soil moisture levels at 50 cm can also shed light on the effects of multi-year drought conditions as well as the anticipated effects on soil drying deeper into the soil profile. Both restoration prescriptions and future monitoring should focus on soil moisture dynamics at 50 cm as an effective target.

The lack of significant forest structure effects on soil moisture at 100 cm underscores our finding that wetter and more stable conditions occur deeper into the soil profile. However, further assessment into the forest structure-driven soil moisture differences at 100 cm reveals nuanced but similar forest density-related patterns as were observed in the shallow 25- and 50-cm depths. Subsetting soil moisture data from sensors in the highest forest density ($n = 2$) and the lowest forest density ($n = 2$) conditions, we show that there are notable differences in soil drying patterns (Figure 10). Specifically, soil moisture at 100 cm in the highest density forest conditions have an earlier soil drying onset, spend more time under the CDT and appear drier overall than in the lowest density conditions. Similar to shallower depths, we assume that the factors contributing to the drier conditions in denser forest include greater rates of snow interception by canopy as well as increased water usage and transpiration rates from the greater number of trees. Importantly, these patterns are observed at opposing ends of the forest-density gradient present at our study site, which reflect heavily-thinned versus overly-dense unthinned forest. While we did not find statistically significant differences among these 100-cm sensors given the limited sample size ($n = 2$ for each end), our results clearly indicate that there is an optimum forest structure that land managers can target to achieve. Our results suggest that this optimum is approximately 30% canopy cover, <100 trees per hectare, and basal area of 25 m^2/ha. Together, these observed trends emphasize the potential benefits of thinning-based restoration on moderating soil drying deep into the soil profile while also pointing to critical unknowns as multi-year drought conditions persist and significant soil drying pushes deeper into the profile.

5 | CONCLUSIONS

We assessed soil moisture response to seasonal drought conditions as well as the impacts of forest structure in a semi-arid forest. Using dense soil water potential time-series data from the top 100 cm of the soil profile, we found significant differences in the timing, magnitude, and overall soil moisture levels across different seasonal drought periods among soil depths and resulting from forest structure conditions. We observed nearly complete drying at all 25 cm locations, a mixture of drying conditions at 50 cm, and a general resistance to drying at 100 cm. Additionally, we found that lower forest density and taller trees were associated with overall wetter soil moisture conditions in soils up to 50 cm deep. Therefore, these results indicate that forest restoration practitioners interested in promoting soil moisture stability focus on continued density reductions and monitoring soil moisture response at 50 cm. A better understanding of these relationships is important given that the continued multi-year drought conditions are likely and the prospect of drying trends translating deeper into the soil profile is strong as drought persists. Future research should untangle the complex relationships between deeper soil moisture and fine-scale forest density and structure. In particular, it should focus on the soil moisture response to the spatial and directional arrangement of trees as well as the size and structural composition of their canopies. This information can help to inform highly effective forest thinning plans that aim to both reduce density and maximize the positive impacts of tree canopy shading and size on soil moisture.

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DATA AVAILABILITY STATEMENT

Data can be made available upon reasonable request from the corresponding author, but is currently not made available due to funding agency restrictions.

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